

Metal–organic frameworks as selectivity regulators for hydrogenation reactions

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Owing to the limited availability of natural sources, the widespread demand of the flavouring, perfume and pharmaceutical industries for unsaturated alcohols is met by producing them from α,β -unsaturated aldehydes, through the selective hydrogenation of the carbon–oxygen group (in preference to the carbon–carbon group)¹. However, developing effective catalysts for this transformation is challenging^{2–7}, because hydrogenation of the carbon–carbon group is thermodynamically favoured⁸. This difficulty is particularly relevant for one major category of heterogeneous catalyst: metal nanoparticles supported on metal oxides. These systems are generally incapable of significantly enhancing the selectivity towards thermodynamically unfavoured reactions, because only the edges of nanoparticles that are in direct contact with the metal–oxide support possess selective catalytic properties; most of the exposed nanoparticle surfaces do not^{9–14}. This has inspired the use of metal–organic frameworks (MOFs) to encapsulate metal nanoparticles within their layers or inside their channels, to influence the activity of the entire nanoparticle surface while maintaining efficient reactant and product transport owing to the porous nature of the material^{15–18}. Here we show that MOFs can also serve as effective selectivity regulators for the hydrogenation of α,β -unsaturated aldehydes. Sandwiching platinum nanoparticles between an inner core and an outer shell composed of an MOF with metal nodes of Fe^{3+} , Cr^{3+} or both (known as MIL-101; refs 19–21) results in stable catalysts that convert a range of α,β -unsaturated aldehydes with high efficiency and with significantly enhanced selectivity towards unsaturated alcohols. Calculations reveal that preferential interaction of MOF metal sites with the carbon–oxygen rather than the carbon–carbon group renders hydrogenation of the former by the embedded platinum nanoparticles a thermodynamically favoured reaction. We anticipate that our basic design strategy will allow the development of other selective heterogeneous catalysts for important yet challenging transformations.

The coordinatively unsaturated metal sites (CUSs) inside MOFs can be readily tuned to adjust interactions between MOFs and reactants, to activate targeted chemical bonds in reactants and thereby to lower the reaction-energy barrier of the desired chemical transformation²². We explore this concept with MOF sandwich nanostructures that are then used as hydrogenation catalysts (Supplementary Fig. 1). The nanostructures contain a layer of platinum (Pt) nanoparticles encapsulated between a core and a shell made of MIL-101, which contains either Fe^{3+} or Cr^{3+} trimers, connected with 1,4-benzenedicarboxylate (BDC) linkers (Fig. 1a and Supplementary Note 1)^{19–21}. All synthesized products are of uniform octahedral shape, with 2.8-nm Pt nanoparticles immobilized between the MIL-101 core and shell (Fig. 1b–l and Supplementary Figs 2–13). Catalytic performance is evaluated using five typical sandwich nanostructures: two MIL-101(Fe)@Pt@MIL-101(Fe) systems with shell thicknesses of about 9.2 nm and 22.0 nm (denoted as MIL-101(Fe)@Pt@MIL-101(Fe)^{9,2} and MIL-101(Fe)@

Pt@MIL-101(Fe)^{22,0}; Fig. 1b–e and Supplementary Figs 8, 9); one MIL-101(Cr)@Pt@MIL-101(Cr) with a shell thickness of around 5.1 nm (denoted as MIL-101(Cr)@Pt@MIL-101(Cr)^{5,1}; Fig. 1f, g and Supplementary Fig. 10); and two MIL-101(Cr)@Pt@MIL-101(Fe) systems with shell thicknesses of around 2.9 nm and 8.8 nm (denoted as MIL-101(Cr)@Pt@MIL-101(Fe)^{2,9} and MIL-101(Cr)@Pt@MIL-101(Fe)^{8,8}; Fig. 1h–l, Supplementary Figs 11–13 and Supplementary Note 2). All of the materials contain single crystalline MIL-101 with a face-centred-cubic (fcc) structure, and a lattice fringe of 2.6 nm that corresponds to the (222) planes of MIL-101 (ref. 19; Fig. 1c–k and Supplementary Figs 14–16). The embedded Pt nanoparticles exhibit high crystallinity, with an interplanar spacing of 0.23 nm that corresponds to the (111) planes of fcc platinum (Fig. 1e, k, insets).

Table 1 summarizes the results obtained when using the sandwich MIL-101@Pt@MIL-101 nanostructures as catalysts for the selective hydrogenation of cinnamaldehyde (A) to cinnamyl alcohol (B), along with results for Pt nanoparticles, MIL-101 and MIL-101@Pt as controls (see also Fig. 1, Supplementary Figs 2–7, and Supplementary Tables 1, 2). Because the C=O and C=C groups in A are both possible hydrogenation targets, we expect a product mixture containing the targeted cinnamyl alcohol (B), as well as hydrocinnamaldehyde (C) and phenyl propanol (D). For a more meaningful evaluation of selectivity for the desired product B, we first compared the catalysts' performances at the same conversion level (about 45%) of A. We found no noticeable hydrogenation of A by MIL-101 (Table 1, entry 9), whereas Pt nanoparticles catalyse this reaction with a turnover frequency (TOF) of 372.4 h^{-1} and a selectivity for B of only around 18.3% (Table 1, entry 8), consistent with density functional theory (DFT) predictions (Supplementary Fig. 17).

The third controls—the MIL-101@Pt nanostructures—exhibit dramatically increased selectivity for B (Table 1, entries 6 and 7), with 86.4% for MIL-101(Fe)@Pt and 44.0% for MIL-101(Cr)@Pt. This remarkable promotion of selective hydrogenation of the C=O bond by MIL-101 is probably due to the CUSs in MIL-101 acting as Lewis acid sites that interact with the C=O bond and activate it²³. A Fourier transform infrared (FTIR) survey shows an obvious redshift of the $\nu_{\text{C=O}}$ bond of A after mixing with MIL-101, while the $\nu_{\text{C=C}}$ bond remains unaltered, confirming a selective interaction between the C=O bond of A and MIL-101 (Supplementary Fig. 18).

DFT calculations with consideration of spin-polarization effect—aimed at understanding the origin of the interaction between MIL-101 and A—show that the five-coordinated metal nodes in the $\text{Fe}_3\text{OCl}(\text{COO})_6\text{H}_2\text{O}$ or $\text{Cr}_3\text{OCl}(\text{COO})_6\text{H}_2\text{O}$ trimers of supertetrahedral MIL-101 cells serve as active sites that interact with the C=O group of A (Supplementary Fig. 19)¹⁹. The calculated adsorption energy of A over the $\text{Fe}_3\text{OCl}(\text{COO})_6\text{H}_2\text{O}$ and $\text{Cr}_3\text{OCl}(\text{COO})_6\text{H}_2\text{O}$ trimers is -1.26 eV and -1.01 eV , respectively (Fig. 2); that is, binding between the trimers and A through Fe–O or Cr–O interactions is thermodynamically favoured, with the interaction between A and

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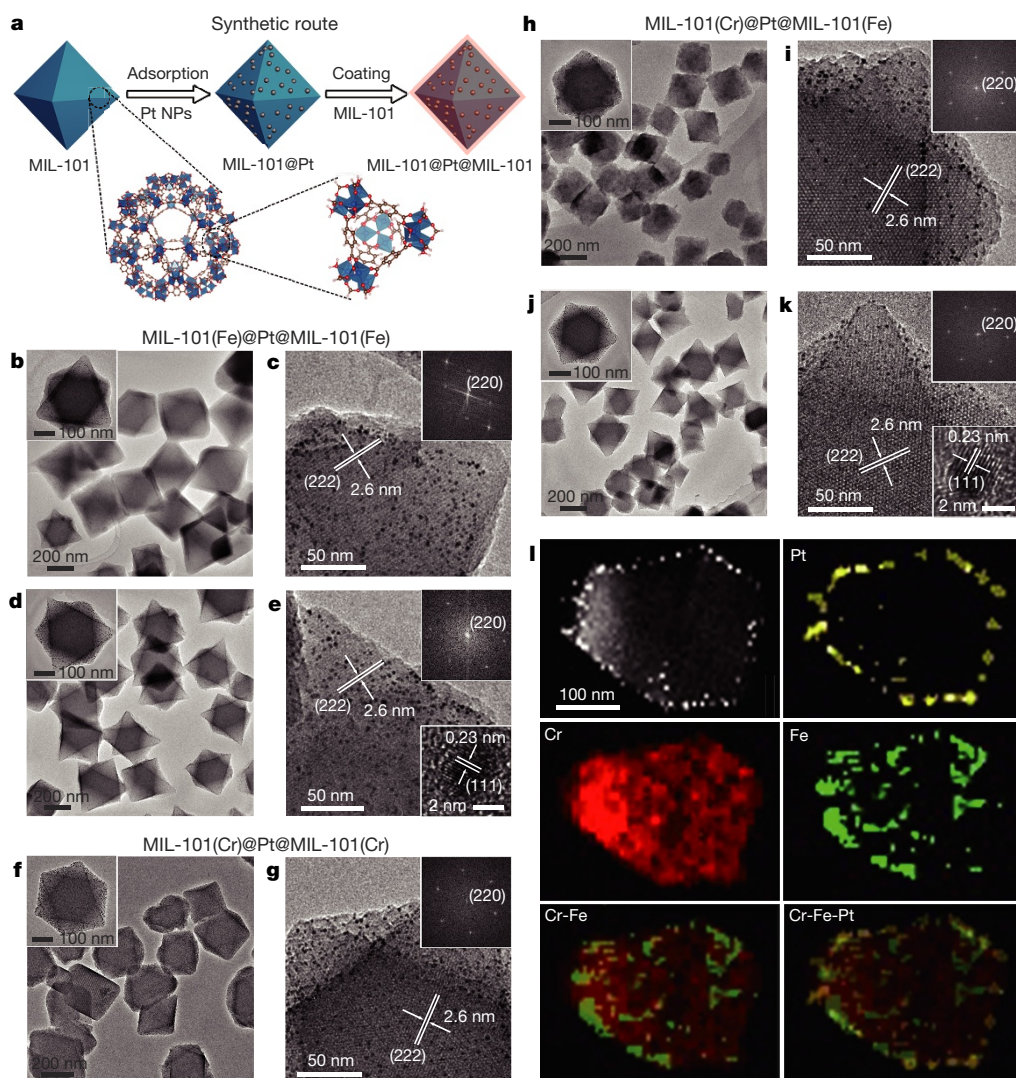


Figure 1 | Sandwich MIL-101@Pt@MIL-101 nanostructures. **a**, Synthetic route to generating sandwich MIL-101@Pt@MIL-101, comprising Pt nanoparticles (NPs) sandwiched between a core and a shell of MIL-101. **b, d**, Transmission electron microscopy (TEM) images of MIL-101(Fe)@Pt@MIL-101(Fe)^{22.0} (**b**) and MIL-101(Fe)@Pt@MIL-101(Fe)^{9.2} (**d**). **c, e**, High-resolution TEM (HR-TEM) images of MIL-101(Fe)@Pt@MIL-101(Fe)^{22.0} (**c**) and MIL-101(Fe)@Pt@MIL-101(Fe)^{9.2} (**e**). **f**, TEM image of MIL-101(Cr)@Pt@MIL-101(Cr)^{5.1}. **g**, HR-TEM image of MIL-101(Cr)@Pt@MIL-101(Cr)^{5.1}. **h, j**, TEM images of MIL-101(Cr)@Pt@MIL-101(Fe)^{8.8} (**h**) and MIL-101(Cr)@Pt@MIL-101(Fe)^{2.9} (**j**). **i, k**, HR-TEM

images of MIL-101(Cr)@Pt@MIL-101(Fe)^{8.8} (**i**) and MIL-101(Fe)@Pt@MIL-101(Fe)^{2.9} (**k**). Insets in **b, d, f, h, j**, TEM images of a single MIL-101@Pt@MIL-101 nanostructure. Insets in **c, e, g, i, k**, corresponding fast Fourier transform (FFT) images of MIL-101@Pt@MIL-101 (top right) and HR-TEM images of Pt NPs (bottom right). **l**, High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) image (top left) and corresponding energy-dispersive X-ray spectroscopy (EDS) elemental mapping images of a cross-section of single MIL-101(Cr)@Pt@MIL-101(Fe)^{2.9}.

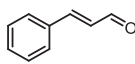
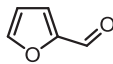
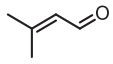
$\text{Fe}_3\text{OCl}(\text{COO})_6\text{H}_2\text{O}$ being the stronger one (Supplementary Figs 20–22 and Supplementary Note 3). The calculated reaction energies for hydrogenation of **A** to **B** or **C** over MIL-101(Fe)@Pt are -2.15 eV and -1.89 eV (Fig. 2a), indicating that the formation of **B** (selectivity approximately 86.4%) is energetically favoured over that of **C** (selectivity roughly 9.6%). In the case of MIL-101(Cr)@Pt, the corresponding reaction energies for **B** and **C** formation are -0.69 eV and -0.92 eV , respectively (Fig. 2b), indicating a smaller difference in the selectivity of **B** (roughly 44.0%) over **C** (about 40.0%). These results confirm the experimental observation that the CUSs in MIL-101 preferentially interact with the $\text{C}=\text{O}$ group (rather than the $\text{C}=\text{C}$ group) of **A** and activate it, giving rise to the improved selectivity for the formation of product **B**.

We also note that at a conversion level of 45%, TOF values are 122.1 h^{-1} for MIL-101(Fe)@Pt and 372.4 h^{-1} for MIL-101(Cr)@Pt (Table 1, entries 6 and 7), indicating that MIL-101(Cr) does not affect

the catalytic activity of Pt nanoparticles whereas MIL-101(Fe) causes a considerable decrease. X-ray photoelectron spectroscopy (XPS) measurements reveal a partial transfer of electrons from the Pt nanoparticles to MIL-101(Fe)²⁴ (Fig. 3a, b, d and Supplementary Note 4), whereas there is no obvious electron transfer between from the Pt nanoparticles to MIL-101(Cr) (Fig. 3a, c, d). This suggests the reduced electron density of Pt nanoparticles in MIL-101(Fe)@Pt as a possible cause of the decreased hydrogenation activity. This is further supported by DFT calculations showing that the negative binding energy of hydrogen atoms (ΔE_{H}) on a platinum(111) (2×2) surface increases with positive charging of the platinum surface (Supplementary Fig. 23), with the resulting stronger surface binding of the hydrogen atoms reducing the catalytic activity of MIL-101(Fe)@Pt²⁵.

Guided by these observations, and given that MIL-101 possesses two characteristic pore windows (of about 1.3 nm and 1.5 nm) that are large enough to accommodate **A** ($1.05\text{ nm} \times 0.65\text{ nm}$; Supplementary

Table 1 | Selective hydrogenation of different α,β -unsaturated aldehydes by different catalysts

Reaction							
$\text{R}-\text{CH}=\text{CH}-\text{CHO} \xrightarrow{\text{H}_2} \text{R}-\text{CH}=\text{CH}-\text{OH} + \text{R}-\text{CH}_2-\text{CHO} + \text{R}-\text{CH}_2-\text{CH}_2-\text{OH}$							
A	B	C	D				
Cinnamaldehyde	Furfural	3-methyl-2-butenal	Acrolein				
							

Entry	Catalysts	Time (h)	Conversion (%) [†]	Selectivity (%) [†]			TOF [‡] (h ⁻¹)
				B	C	D	
Substrate: cinnamaldehyde*							
1	MIL-101(Fe)@Pt@MIL-101(Fe) ^{9,2}	8.5	45.0	94.1	2.9	3.0	18.0
2	MIL-101(Fe)@Pt@MIL-101(Fe) ^{22,0}	10	45.0	96.3	1.9	1.8	15.3
3	MIL-101(Cr)@Pt@MIL-101(Fe) ^{2,9}	2	45.0	94.2	4.7	1.1	76.3
4	MIL-101(Cr)@Pt@MIL-101(Fe) ^{8,8}	6	45.0	94.5	2.8	2.7	25.4
5	MIL-101(Cr)@Pt@MIL-101(Cr) ^{5,1}	0.7	45.0	79.2	14.6	6.2	218.1
6	MIL-101(Fe)@Pt	1.25	45.0	86.4	9.6	4.0	122.1
7	MIL-101(Cr)@Pt	0.41	45.0	44.0	40.0	16.0	372.4
8	Pt NPs	0.41	45.0	18.3	60.6	21.1	372.4
9	MIL-101(Cr) or MIL-101(Fe)	24	0	—	—	—	0
10	MIL-101(Fe)@Pt@MIL-101(Fe) ^{9,2}	24	94.3	97.0	0	3.0	13.3
11	MIL-101(Fe)@Pt@MIL-101(Fe) ^{22,0}	24	86.3	97.4	0.3	2.3	12.2
12	MIL-101(Cr)@Pt@MIL-101(Fe) ^{2,9}	20	99.8	95.6	0.8	3.6	16.9
13	MIL-101(Cr)@Pt@MIL-101(Fe) ^{8,8}	20	90.6	94.3	2.1	3.6	15.4
14	MIL-101(Cr)@Pt@MIL-101(Cr) ^{5,1}	2	95.1	62.6	12.2	25.2	161.3
Substrate: furfural*							
15	MIL-101(Fe)@Pt@MIL-101(Fe) ^{9,2}	15	85.6	93.2	3.5	3.3	19.4
16	MIL-101(Cr)@Pt@MIL-101(Fe) ^{8,8}	7	87.9	96.5	3.5	0	42.6
17	MIL-101(Cr)@Pt@MIL-101(Cr) ^{5,1}	5	98.5	99.8	0	0.2	66.8
Substrate: 3-methyl-2-butenal*							
18	MIL-101(Fe)@Pt@MIL-101(Fe) ^{9,2}	17	71.1	87.8	3.2	9.0	14.2
19	MIL-101(Fe)@Pt@MIL-101(Fe) ^{22,0}	24	59.9	92.5	2.8	4.7	8.5
20	MIL-101(Cr)@Pt@MIL-101(Fe) ^{8,8}	10	85.0	64.4	7.9	27.7	28.8
Substrate: acrolein*							
21	MIL-101(Fe)@Pt@MIL-101(Fe) ^{9,2}	3	76.9	75.7	19.0	5.3	86.9
22	MIL-101(Fe)@Pt@MIL-101(Fe) ^{22,0}	3	52.7	97.3	2.7	0	59.6
23	MIL-101(Cr)@Pt@MIL-101(Fe) ^{8,8}	0.75	87.5	57.9	17.3	24.8	395.8
24	Au ₂₅ (SR) ₁₈ /Fe ₂ O ₃ [¶]	3	47.0	92.0	—	—	3.1
25	Au clusters coated with <i>tert</i> -butyl (naphthalen-1-yl)phosphine oxide [§]	18	61.0	>99	—	—	1.7

A, α,β -unsaturated aldehydes (cinnamaldehyde, furfural, 3-methyl-2-butenal or acrolein); **B**, unsaturated alcohol; **C**, saturated aldehyde; **D**, saturated alcohol.

*Reaction condition: each catalyst contains 0.23 mg Pt NPs; the reaction requires 0.4 mmol of **A**, room temperature and 3 MPa H₂.

[†]The percentage conversion of **A** and selectivity for specific products were determined by gas chromatography-mass spectrometry and gas chromatography or ¹H NMR.

[‡]TOF was calculated by the mole number of converted **A** (mole number of total Pt or Au)⁻¹ h⁻¹.

[¶]Reaction condition²⁶: 100 mg 1%Au₂₅(SR)₁₈/Fe₂O₃, 0.1 mmol **A**, 0 °C and 0.1 MPa H₂. The catalytic selective hydrogenation of acrolein has been described previously.

[§]Reaction condition²⁷: 0.01 mmol Au clusters (assuming 46.28% Au on the basis of element analysis), 0.5 mmol **A**, 5 ml tetrahydrofuran, 40 °C and 4 MPa H₂. The catalytic selective hydrogenation of acrolein has been described previously.

Table 2 and Figs 24, 25), we next explored the sandwich MIL-101@Pt@MIL-101 structure as an ideal catalyst for selective hydrogenation of **A** to **B**. Entries 1–5 in Table 1 confirm that coating with MIL-101 shells of different thicknesses considerably enhances the selectivity for **B**: the selectivity with MIL-101(Cr)@Pt@MIL-101(Cr)^{5,1} is 79.2%, while using MIL-101(Cr) or MIL-101(Fe) cores that are coated with MIL-101(Fe) shells gives selectivities that are always higher than 94%. The improved selectivities are accompanied by a decrease in TOF values, which is expected because the MIL-101 shells slow down reactant and product diffusion. Coating with MIL-101(Fe) leads to an additional decrease in catalytic activity, owing to the interfacial electron-transfer effect (Fig. 3 and Supplementary Fig. 23).

Our MIL-101@Pt@MIL-101 catalysts combine exceptionally high selectivity and conversion efficiency (Table 1, entries 10–14, and Supplementary Table 3), and outperform state-of-the-art catalysts (Supplementary Table 4). MIL-101(Cr)@Pt@MIL-101(Fe)^{2,9}, for example, exhibits excellent selectivity (95.6%) and almost full conversion

(99.8%) (Supplementary Fig. 26). Moreover, reusability tests indicate that for MIL-101(Cr)@Pt@MIL-101(Fe)^{2,9}, both the conversion efficiency of **A** and the selectivity for **B** remain almost unchanged over five successive catalytic cycles (Supplementary Figs 26–30, 31a), with the stability of the system being further verified by XRD and transmission electron microscope (TEM) measurements (Supplementary Figs 31b, 32) that show no evidence for structural or morphological differences between fresh and used catalysts. Excellent catalytic stability is also obtained with MIL-101(Fe)@Pt@MIL-101(Fe)^{9,2} (Supplementary Fig. 33).

To further confirm that the integration of Pt nanoparticles and CUSs into a single sandwich nanostructure results in useful selective hydrogenation capabilities, we used MIL-101@Pt@MIL-101 to hydrogenate smaller-sized α,β -unsaturated aldehydes. Acrolein (0.69 nm × 0.51 nm), with no substituents on C=C bond, branched 3-methyl-2-butenal (0.79 nm × 0.60 nm), and furfural (0.81 nm × 0.64 nm), with a furan ring (see Supplementary Fig. 34 for

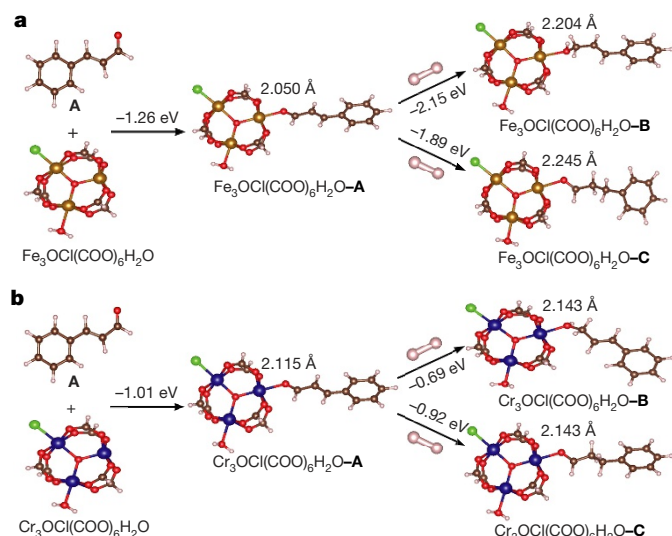


Figure 2 | Investigation of the reaction mechanism on MIL-101@Pt. a, b, Theoretical calculations of the interaction of cinnamaldehyde (A) with Fe trimers (a) or Cr trimers (b), and the subsequent selective hydrogenation of A to cinnamyl alcohol (B) and hydrocinnamaldehyde (C), on a MIL-101(Fe)@Pt or MIL-101(Cr)@Pt catalyst. Brown, carbon; pink, hydrogen; red, oxygen; gold, iron; blue, chromium; green, chlorine.

structures), are all converted through preferential hydrogenation of the C=O group over the C=C group (Table 1, Supplementary Tables 5–7 and Supplementary Figs 35–37). Conversion efficiencies are 52.7%, 59.9% and 98.5%, respectively, with selectivities for hydrogenation of the C=O group being 97.3%, 92.5% and 99.8%. When compared with other catalysts^{26,27}, MIL-101(Fe)@Pt/MIL-101(Fe)^{22,0} also exhibits excellent selectivity towards allyl alcohol (Table 1, entries 24 and 25; Supplementary Note 5 and Supplementary Table 8).

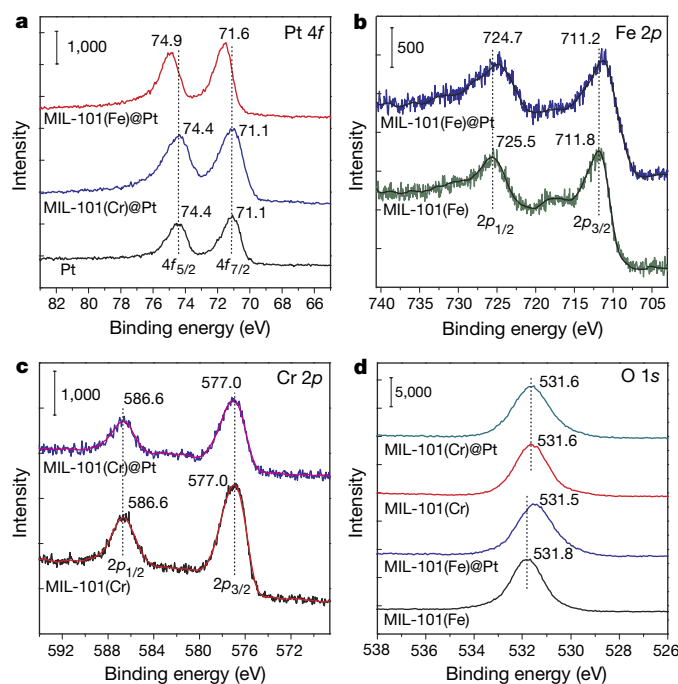


Figure 3 | XPS profiles of Pt, Fe, Cr and O in different catalysts. a, Pt 4f level. b, Fe 2p level. c, Cr 2p level. d, O 1s level. The catalysts investigated were pure Pt NPs, MIL-101(Fe), MIL-101(Cr), MIL-101(Cr)@Pt and MIL-101(Fe)@Pt. Numbers adjacent to dashed lines represent the peaks' binding energy values.

To explore the generality of the catalyst preparation method, we also synthesized^{28–30} the sandwich structures MOF-525(Zr)@Pt@MOF-525(Zr)^{26,5}, MOF-74(Co)@Pt@MOF-74(Co)^{8,4}, UiO-66(Zr)@Pt@UiO-66(Zr)^{11,2}, and UiO-67(Zr)@Pt@UiO-67(Zr)^{24,1} (Supplementary Table 2 and Supplementary Figs 38–45), which catalysed the hydrogenation of A with conversion efficiencies of 35.2%, 35.0%, 90.0% and 69.4%, respectively, and selectivities towards B of 85.0%, 70.1%, 65.0% and 73.0%, respectively (Supplementary Figs 46–49 and Supplementary Table 9). Finally, as an example of a system containing different metal nanoparticles, we also prepared MIL-101(Fe)@Ru@MIL-101(Fe)^{8,3}, which hydrogenated A with a conversion efficiency of 48.7% and a selectivity towards B of 58.5% (Supplementary Figs 50–52 and Supplementary Table 10). These results illustrate the generality of the concept of using CUSs in MOFs as a means of tuning the selectivity of the hydrogenation of α,β -unsaturated aldehydes. Moreover, using commercial Pt/C (carbon) and Pt/Fe₂O₃ for the hydrogenation of A (Supplementary Figs 53, 54) resulted in conversion efficiencies of 98.1% and 54.5% and selectivities towards B of 39.9% and 84.5%, respectively (Supplementary Table 11), with the different selectivities suggesting that Fe-based supports favour the hydrogenation of C=O groups over C=C groups (although not to the extent seen with the sandwich forms of catalysts). Taken together, our observations confirm the considerable potential of MOFs as a new generation of heterogeneous catalytic supports that may prove effective when targeting important but highly challenging reactions.

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Author Contributions Z.T. and G.L. proposed the research direction and guided the project. M.Z., K.Y. and G.L. designed and performed the materials synthesis, characterization, and catalytic tests. Y.W. and H.Z. performed the theoretical calculations and explained the catalytic results. J.G. helped to detect and analyse the Brunauer–Emmett–Teller (BET) surface areas and pore-size distributions of samples. L.G. guided the high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) imaging of sandwich structures. W.H. took part in the characterization of some samples, and discussed the results. Z.T., G.L., M.Z. and Y.W. drafted the manuscript. All of the authors discussed the results and commented on the manuscript.

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